

Induced emf and Current

PROBLEM

A coil with 25 turns of wire is wrapped around a hollow tube with an area of 1.8 m^2 . Each turn has the same area as the tube. A uniform magnetic field is applied at a right angle to the plane of the coil. If the field increases uniformly from 0.00 T to 0.55 T in 0.85 s , find the magnitude of the induced emf in the coil. If the resistance in the coil is 2.5Ω , find the magnitude of the induced current in the coil.